

One-Way ANOVA and Post-Hoc Tests

ITAO 20200: Foundations of Statistics

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Description

Analysis of Variance (ANOVA) is one of the most versatile statistics ever developed, and we will see it in a few different places in this course. Its main attribute is the ability to parse out the variability components into what is due to various treatment effects versus what is due to sampling error. More specifically, the one-way ANOVA is used to compare the means of three or more groups simultaneously without worry of inflating Type I Error rates.

Assumptions

Only one independent and one dependent variable should be used. The independent variable (the grouping variable, or factor) can have any number of levels, but the levels should be independent of one another; in other words, respondents cannot be included in more than one category (else a repeated-measures ANOVA needs to be performed). The dependent variable should be measured on an interval or ratio scale. The scores on the dependent variable should be normally distributed, and the variances of the populations from which the samples came should be equal.

Data Entry

For this test, two separate variable columns must be created, one for each observation. For this test, we will use the `Grades.csv` data. (This data set was also used in the Paired-Samples/Dependent t Test instruction sheet.) In this data set, there are five variables. **Pretest**, **Midterm**, and **Final** all represent scores obtained on those tests. **Teacher** is a grouping variable indicating which of three different instructors (1 = *Dr. NiceGuy*, 2 = *Dr. Funny*, 3 = *Dr. Genius*) taught the students. **Required** is another grouping variable indicating whether or not the class was an *elective* (0) or was *required* (1).

In our example, we will use **Teacher** as our factor (or independent variable) and **Final** as our dependent variable. This will allow us to determine if the instructor has any effect on the final exam grades in the course. A portion of the data is given below.

```
Grades <- read.csv("https://mendoza-stats.s3.amazonaws.com/grades.csv")
head(Grades, n = 3) # First 3 observations
```

##	Pretest	Midterm	Final	Teacher	Required
## 1	56	75	77	1	1
## 2	79	91	84	1	0
## 3	68	82	81	1	1

```
tail(Grades, n = 3) # Last 3 observations
```

```
##      Pretest Midterm Final Teacher Required
## 22      61      83    94      3      0
## 23      77      95    92      3      1
## 24      69      84    92      3      0
```

Data Analysis and Output

Before proceeding with our ANOVA, let's transform our grouping variable into a factor with 3 identifiable levels.

```
Grades$Teacher <- factor(Grades$Teacher, levels = c(1, 2, 3),
                        labels = c("Dr. NiceGuy", "Dr. Funny", "Dr. Genius"))
```

After the transformation, this is how our data should look like:

```
head(Grades, n = 3) # First 3 observations
```

```
##      Pretest Midterm Final      Teacher Required
## 1      56      75    77 Dr. NiceGuy      1
## 2      79      91    84 Dr. NiceGuy      0
## 3      68      82    81 Dr. NiceGuy      1
```

```
tail(Grades, n = 3) # Last 3 observations
```

```
##      Pretest Midterm Final      Teacher Required
## 22      61      83    94 Dr. Genius      0
## 23      77      95    92 Dr. Genius      1
## 24      69      84    92 Dr. Genius      0
```

Some Basic Descriptive Statistics

Now, let's obtain some basic descriptive statistics based on the labeled groups. Since our analysis focuses on group differences, it is important that the summary statistics reflect values for each group. One way to separate final exam scores based on instructor groups is to use the `by` command. Individual commands can be run for specific summary statistics, such as group means, standard deviations, and sample sizes.

```
by(Grades$Final, Grades$Teacher, mean) #Dependent var, Independent var, statistic
```

```
## Grades$Teacher: Dr. NiceGuy
## [1] 79
## -----
## Grades$Teacher: Dr. Funny
## [1] 84.125
## -----
## Grades$Teacher: Dr. Genius
## [1] 90.75
```

```

by(Grades$Final, Grades$Teacher, sd)      #Dependent var, Independent var, statistic

## Grades$Teacher: Dr. NiceGuy
## [1] 4.780914
## -----
## Grades$Teacher: Dr. Funny
## [1] 8.871101
## -----
## Grades$Teacher: Dr. Genius
## [1] 6.964194

by(Grades$Final, Grades$Teacher, length)  #Dependent var, Independent var, statistic

## Grades$Teacher: Dr. NiceGuy
## [1] 8
## -----
## Grades$Teacher: Dr. Funny
## [1] 8
## -----
## Grades$Teacher: Dr. Genius
## [1] 8

```

More Advanced Code for Summary Statistics Rather than run individual lines of code reflecting specific descriptive statistics, some might find it preferable to write one block of code that calculates numerous statistics simultaneously. The following block of code calculates a set of summary statistics for final exam scores, broken down separately by instructor. The `by()` function is still the key command here: it splits the `Final` column into groups based on the categories in the `Teacher` column (Dr. NiceGuy, Dr. Funny, and Dr. Genius), and then applies a custom calculation to each group separately.

The custom calculation is defined inside the `function(x)` part of the code, where `x` stands in for the final exam scores within each instructor group. For each group, it computes seven statistics: the number of students in that group (`n`), the average score (`mean`), the standard deviation (`sd`), the median, the minimum and maximum score in the group (`min` and `max`), and the standard error of the mean (`se`). Most of these values are rounded to two decimal places for readability.

The end result is a side-by-side comparison; for each instructor, you get a compact summary of how many students there are, what their typical score looks like, how much scores vary, and the range of scores observed. This makes it easy to compare performance patterns between groups at a glance.

```

by(Grades$Final, Grades$Teacher, function(x) {
  c(n      = length(x),
    mean   = round(mean(x), 2),
    sd     = round(sd(x), 2),
    median = round(median(x), 2),
    min    = min(x),
    max    = max(x),

```

```
se      = round(sd(x) / sqrt(length(x)), 2))
})
```

```
## Grades$Teacher: Dr. NiceGuy
##      n   mean    sd median   min    max    se
##  8.00  79.00  4.78  79.00  72.00  86.00  1.69
## -----
## Grades$Teacher: Dr. Funny
##      n   mean    sd median   min    max    se
##  8.00  84.12  8.87  86.00  69.00  95.00  3.14
## -----
## Grades$Teacher: Dr. Genius
##      n   mean    sd median   min    max    se
##  8.00  90.75  6.96  92.00  79.00 100.00  2.46
```

One-Way ANOVA Test

Next, let's perform our ANOVA test. We wish to test the hypotheses:

$$H_0 : \mu_1 = \mu_2 = \mu_3$$

$$H_a : \text{at least one } \mu_i \text{ is different}$$

For this test, we must identify the dependent and independent variables. We will be using the `aov()` function in R. The first argument is our dependent variable, and the second argument is the independent variable. For our data, the `aov()` function is utilized as follows:

```
# One-way ANOVA for Final scores and Teacher
Grades.ANOVA <- aov(Grades$Final ~ Grades$Teacher) # Save results into an object
summary(Grades.ANOVA) # summary() function displays the results of the analysis
```

```
##              Df Sum Sq Mean Sq F value Pr(>F)
## Grades$Teacher  2  555.3  277.63    5.551 0.0116 *
## Residuals      21 1050.4   50.02
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

The source table should be interpreted in the same way that we discussed in lecture, noting that what our text calls “Treatment” is named the independent variable in R (in this case, **Teacher**), and what our text calls “Error”, R calls **Residuals**.

Interpreting the ANOVA Source Table

The **F value** column gives us the obtained test statistic, and the **Pr(>F)** column indicates the *exact one-tailed p-value* of our obtained test statistic. If this *p*-value is less than or equal to our alpha level (0.05 here), then we have determined that there is a statistically significant difference between our groups. If this

p -value is greater than our alpha level, then we retain the null hypothesis and state that the evidence is inconclusive. *If the null hypothesis is retained, then there is no reason to continue to process post-hoc information.*

Post-Hoc Analysis

Our one-way ANOVA produced a statistically significant result. Therefore, it is appropriate to investigate further with post-hoc testing. There are many ways in which post-hoc testing can be done. We will be using **pairwise comparison** post-hoc procedures in this class, meaning that we will make comparisons between two group means at a time to determine if they are statistically different from one another. For our post-hoc testing, our null hypothesis suggests that there is no difference between means being evaluated ($H_0 : \mu_a = \mu_b$), or that the difference between those means would be zero ($H_0 : \mu_a - \mu_b = 0$).

Tukey's HSD

Tukey's HSD (Honestly Significant Difference) is one method of determining which groups are significantly different from one another after an omnibus F test has been rejected. The HSD is the minimum value that needs to be observed between two means before we say those groups are statistically and significantly different. R does not report this minimum value, but it uses the minimum value in its determination of effects.

We will use the `TukeyHSD()` function in R for this post-hoc testing. The `TukeyHSD()` takes the fitted ANOVA as an argument and it is utilized as follows:

```
TukeyHSD(Grades.ANOVA,      # Takes the fitted ANOVA as an argument
          conf.level = 0.95) # confidence level = 1 - alpha
```

```
## Tukey multiple comparisons of means
## 95% family-wise confidence level
##
## Fit: aov(formula = Grades$Final ~ Grades$Teacher)
##
## $`Grades$Teacher`
##              diff      lwr      upr    p adj
## Dr. Funny-Dr. NiceGuy  5.125 -3.788153 14.03815 0.3348406
## Dr. Genius-Dr. NiceGuy 11.750  2.836847 20.66315 0.0086949
## Dr. Genius-Dr. Funny   6.625 -2.288153 15.53815 0.1711222
```

The results provide us with the mean difference for each pairing with corresponding confidence interval and Tukey adjusted p -values. The `diff` column provides the mean difference between the stated groups. The `lwr` and `upr` columns provide the lower and upper bound estimates of the confidence interval (respectively) for the mean difference between those groups. The `p adj` provides the Tukey adjusted p -value for each pairwise comparison.

There are two ways of determining if the differences are statistically significant from this output. First, if the confidence interval contains zero (the null value), then it is a possible true mean difference for this pairwise

comparison. As such, the respective two group means are not significantly different. If zero is outside of the confidence interval, then we can conclude that the groups are significantly different. Alternatively, we can use the adjusted p -values for decision making. If the adjusted p -value is less than or equal to our chosen experimentwise alpha level (here, 0.05), then the respective two group means are statistically different.

For instance, the average final exam score for Dr. Genius is significantly different from that of Dr. NiceGuy. We can tell that the sample mean for Dr. NiceGuy is lower than the sample mean for Dr. Genius because the `diff` value is positive. The mean difference between Dr. NiceGuy and Dr. Genius is 11.750. The confidence interval surrounding this mean difference extends from 2.837 to 20.663. Zero is not included in this interval, and thus it is not a likely value for the true difference between these means. Alternatively, the adjusted p -value for this comparison is 0.009 (rounded), which is less than our chosen experimentwise alpha level of 0.05. This would also lead us to reject the null hypothesis.

Unadjusted Fisher's LSD

An unadjusted Fisher's LSD (Least Significant Difference) can be used to determine which groups are significantly different from one another after an omnibus F test has been rejected. The LSD is also the minimum value that needs to be observed between two means before we say those groups are statistically significantly different. ***** However, since no adjustments have been made, this post-hoc test grossly inflates the Type I Error rate if multiple comparisons are sought and the results are evaluated as shown. Therefore, an output adjustment must be performed before using these results.*****

We will use the `pairwise.t.test()` function in R for this post-hoc testing. The `pairwise.t.test()` takes a response variable (dependent variable) and the grouping variable or factor (independent variable) as arguments.

Using the unadjusted LSD for our data we have:

```
pairwise.t.test(Grades$Final,           # dependent variable
                Grades$Teacher,         # independent variable
                p.adjust.method = "none", # "none" means unadjusted LSD
                alternative = "two.sided")
```

```
##
## Pairwise comparisons using t tests with pooled SD
##
## data:  Grades$Final and Grades$Teacher
##
##           Dr. NiceGuy Dr. Funny
## Dr. Funny  0.1620      -
## Dr. Genius 0.0032      0.0750
##
## P value adjustment method: none
```

The results are given in terms of a matrix of p -values. In order to interpret the output correctly from an unadjusted LSD, **you should multiply the p -values by the overall number of multiple comparisons made in your study.** In this case, up to three pairwise comparisons can be made, but not all need be.

Thus, each p -value is utilized by itself (see below) or is multiplied by 2 or 3 depending on how many comparisons are sought. Once you have done that, you can compare your overall results against your alpha level (here, 0.05).

Based on this output, we have:

	Unadjusted p -value (okay to use for 1 intended post-hoc pairwise comparison)	Adjusted p -values for 2 pairwise comparison	Adjusted p -values for 3 pairwise comparison
Dr. NiceGuy vs Dr. Funny	0.1620	0.3240	0.4860
Dr. NiceGuy vs Dr. Genius	0.0032*	0.0064*	0.0096*
Dr. Funny vs Dr. Genius	0.0750	0.1500	0.2250

Regardless of the number of intended pairwise comparisons, there is only one statistically significant mean difference even after our adjustments: Dr. NiceGuy and Dr. Genius. There are no other statistically significant differences between the groups.

Bonferroni Adjusted LSD

The Bonferroni adjusted LSD (Least Significant Difference) is also calculated to determine which groups are significantly different from one another after an omnibus F test has been rejected. This test **has been adjusted for all possible pairwise comparisons**, and therefore the output can be used as it is to indicate which groups differ significantly from one another. As noted previously, the LSD is the minimum value that needs to be observed between two means before we say those groups are statistically significantly different. R does not report the calculated LSD value, but it does provide the p -values to be utilized for making a decision based on our alpha level (here, 0.05). We will utilize the `pairwise.t.test()` function again, but with only one modification for the Bonferroni adjusted LSD as follows:

```
pairwise.t.test(Grades$Final,           # dependent variable
                Grades$Teacher,         # independent variable
                p.adjust.method = "bonferroni", # Bonferroni adjusted LSD
                alternative = "two.sided")
```

```
##
## Pairwise comparisons using t tests with pooled SD
##
## data: Grades$Final and Grades$Teacher
##
##           Dr. NiceGuy Dr. Funny
## Dr. Funny 0.4861      -
## Dr. Genius 0.0097      0.2250
##
## P value adjustment method: bonferroni
```

Similar to the unadjusted LSD, the results are given as a matrix of the Bonferroni adjusted p -values. To determine if the differences are statistically significant, compare the adjusted p -values to the experimentwise alpha level. If the adjusted p -value is less than or equal to our chosen alpha level, then the respective two group means are statistically different.

Look at the comparison between Dr. NiceGuy and Dr. Genius. The average final exam score for Dr. NiceGuy is significantly different from that of Dr. Genius, with Dr. Genius's students earning the higher average. The p -value for this comparison is 0.0097, which is less than our chosen alpha level of 0.05. (**Note:** The p -values reported in the Bonferroni adjusted section match (within rounding error) the adjusted p -values from the LSD section above when all pairwise comparisons are made.)

Writing Your Results

To convey your results properly, you need to know the F value, the degrees of freedom, and the p -value of the test statistic. You should also have at least one of the post-hoc tests at your disposal. It would be helpful to have the mean and standard deviation of the groups as well. Though multiple approaches were used to demonstrate the code and output, only one approach (Tukey's HSD) is explained below.

A one way analysis of variance was conducted to determine if final exam scores were influenced by instructor, and the results were statistically significant, ($F(2, 21) = 5.551$, $p = 0.0116$). Tukey's HSD was used to probe the significant effect. It was determined that the average final exam grade for students who took the class with Dr. NiceGuy ($\bar{x} = 79.00$, $s = 4.78$) was significantly lower than the average grade for students who took the class with Dr. Genius ($\bar{x} = 90.75$, $s = 6.96$). No significant differences were observed between either of these two professors and Dr. Funny ($\bar{x} = 84.12$, $s = 8.87$).

Practice on Your Own

Read the file called `paintdrying.csv` into R. This data is a modified version of problem 11 from chapter 13. Four different paints are advertised as having the same drying time. To check the manufacturers' claims, five samples were tested for each of the paints. The time (in minutes) until the paint was dry enough to apply a second coat was recorded. Using $\alpha = 0.05$, determine whether or not the mean drying time is the same for each brand of paint.

Use the following code to import the dataset into R:

```
PaintDrying <- read.csv("https://mendoza-stats.s3.amazonaws.com/paintdrying.csv")
```

1. Find and complete the summary statistics table below:

Paint	Sample Size, n	Sample Mean, $\bar{x}$	Sample Std Dev, s
Paint 1			
Paint 2			
Paint 3			
Paint 4			

2. What are the null and alternative hypotheses for this test?

3. Generate the ANOVA source table.
4. What are your decision and conclusion for this test? Does the manufacturer of a paint have an effect on the drying times? Write your results in proper statistical notation.
5. Is a post-hoc analysis necessary? If yes,
 - a. conduct Tukey's HSD and Bonferroni adjusted LSD tests.
 - b. compare your results from the two post-hoc tests.
 - c. what conclusion can you draw from the post-hoc tests.